

Intrinsic Token Weighting for Credit Assignment in Language Model Reinforcement Learning

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Abstract

Policy gradient methods for LLM reinforcement learning rely on value function critics, but language generation prefixes are non-Markov and poorly suited for value estimation, often introducing bias. Recent critic-free approaches such as GRPO and RLOO eliminate value networks but still assign trajectory rewards uniformly across tokens, leaving the credit assignment problem unresolved. In this work, we study token-level credit redistribution without value functions, using intrinsic token-weighting schemes derived from the model’s predictive distribution, including surprisal- and entropy-based weights. These methods require no additional networks, no tree construction, and no reward models. Through experiments on verifiable math and code tasks, we show that intrinsic weighting reduces gradient variance, stabilizes training, and yields interpretable credit assignment while matching established critic-free baselines. Our results demonstrate that redistributing credit via intrinsic distributional signals is a viable alternative to advantage estimation in language RL.

1 Introduction

Reinforcement learning has become an important component of large language model (LLM) training pipelines, particularly for alignment, preference optimization, and reasoning tasks. Most contemporary approaches rely on policy gradient methods augmented with advantage estimation, typically implemented via learned value functions or generalized advantage estimation. These techniques originate from classical reinforcement learning settings in which states are well-defined, approximately Markov, and admit stable value estimates.

Applying this framework to language generation introduces a fundamental mismatch. In autoregressive LLMs, the ‘state’ is a partial sequence of tokens. Such prefixes are not Markov, do not summarize progress toward task completion, and can vary widely in semantic meaning despite superficial similarity. Consequently, estimating a value function over prefixes is both statistically challenging and conceptually ill-defined. Empirically, value heads in language RL are often brittle, sensitive to distributional shift, and prone to introducing bias that outweighs their variance-reduction benefits.

Motivated by these issues, recent work has increasingly moved away from learned critics in LLM reinforcement learning. Critic-free, outcome-based methods such as RLOO and GRPO replace value estimation with batch-level baselines, achieving stable training on verifiable reasoning tasks such as math and code. However, these approaches still implicitly assume uniform credit assignment across tokens: every action in a trajectory receives the same reward signal, modulated only by a scalar baseline.

This raises a natural question: *if we no longer estimate values over prefixes, how should trajectory-level reward be distributed across tokens?* Uniform credit assignment is simple, but it ignores the fact that different tokens play very different roles in generation. Some tokens correspond to high-uncertainty decisions, strategy commitments, or branching points in reasoning, while others are low-entropy continuations or stylistic filler. Treating all tokens equally risks diluting the learning signal and obscuring the mechanisms by which policies improve.

In this work, we study token-level credit redistribution as a mechanism for on-policy LLM reinforcement

learning without value functions. Rather than estimating advantages or prefix values, we propose to weight token-level policy gradients using intrinsic signals derived directly from the model’s predictive distribution. These include token surprisal and changes in predictive entropy, which naturally reflect decision difficulty and commitment during generation. Importantly, these signals are inexpensive to compute, require no additional networks or tree structures, and make no assumptions about the existence of a meaningful value function over prefixes.

We conduct a systematic empirical evaluation of intrinsic token-weighting schemes on verifiable math and code reasoning tasks. We compare uniform outcome-based policy gradients, batch-baseline methods such as RLOO/GRPO, and several intrinsic weighting strategies under otherwise identical training conditions. Our results show that intrinsic token weighting can reduce gradient variance, stabilize optimization, and produce interpretable credit assignment patterns, while achieving performance comparable to established critic-free baselines.

Overall, this paper reframes advantage estimation as one instantiation of a broader credit assignment problem in language reinforcement learning. We demonstrate that, in settings where state semantics are weak and rewards are trajectory-level, redistributing credit using intrinsic distributional signals is a practical and effective alternative to value-based methods.

2 Related Work

2.1 From RLHF to RLVR

Modern language-model post-training inherits its optimization machinery from policy-gradient reinforcement learning, from REINFORCE [25] to trust-region and clipped-surrogate methods such as TRPO and PPO [16, 18]. Advantage estimation, especially GAE, became the standard variance-reduction device in continuous-control RL by learning value functions over states or prefixes [17]. In language modeling, these ideas entered mainstream use through RLHF systems that optimize sequence-level rewards derived from human preferences or reward models [5, 21, 14].

The recent reasoning-model wave shifted part of the field from preference-based RLHF toward reinforcement learning with verifiable rewards (RLVR), where supervision is often a deterministic correctness signal on math or code tasks. DeepSeekMath introduced GRPO as a practical critic-free alternative for these settings [19], and DeepSeek-R1 popularized large-scale RL-only or RL-dominant reasoning pipelines [7]. Subsequent surveys document how quickly this regime became the default template for open reasoning-model replication and extension [29]. This transition matters for our setting because RLVR makes sparse outcome rewards and long reasoning trajectories central, thereby exposing the credit-assignment problem more directly than earlier short-form RLHF tasks.

2.2 Critic-Based and Critic-Free Policy Optimization

The classical solution to delayed reward is to learn a critic or value function and convert returns into token- or prefix-level advantages. PPO-style RLHF pipelines follow this recipe, but the value-estimation problem is particularly awkward for text generation because prefixes are high-dimensional, non-Markov, and semantically heterogeneous. A growing body of work therefore revisits whether value heads are necessary at all in LLM post-training.

ReMax replaces learned critics with a greedy baseline and shows that much of PPO’s practical benefit can be retained with a simpler REINFORCE-style objective [12]. Back to Basics systematically compares PPO with RLOO-style estimators and finds that critic-free methods can match or exceed value-based baselines in RLHF [1]. REINFORCE++ further strengthens this line by improving robustness to prompt and reward-model variation without reintroducing a learned value function [9]. In reasoning-focused RLVR, GRPO normalizes rewards across a group of sampled responses rather than learning prefix values [19], and later work refines this family with theoretical analyses, off-policy variants, and sequence-level formulations such as GSPO [13, 30]. These methods substantially simplify the optimization stack, but they generally still broadcast a trajectory-level signal across all tokens once a scalar baseline is fixed.

2.3 Reasoning-Specific Analyses of RLVR

Another recent line asks why critic-free RLVR works so well for reasoning in the first place. A Minimalist Approach to LLM Reasoning argues that strong reasoning gains can emerge from a comparatively simple recipe that moves from rejection sampling to REINFORCE-style online updates, suggesting that some of the field’s complexity is optional rather than essential [26]. Complementarily, Wen et al. analyze RLVR theoretically and empirically, arguing that answer-level verifiable rewards can nonetheless incentivize correct intermediate reasoning early in a trajectory [24]. These results are important because they show that uniform outcome-level objectives already contain nontrivial reasoning signal.

At the same time, they do not eliminate the question of *which* tokens should absorb that signal. Recent empirical analysis of RLVR training dynamics suggests that improvements are disproportionately driven by a minority of high-entropy “forking” tokens, and that restricting updates to this subset can remain competitive or even outperform full-token updates in some settings [23]. This is the closest recent line to our work conceptually: it supports the claim that token salience is highly non-uniform during reasoning RL. Our work differs in scope and mechanism, however, by studying general intrinsic weighting schemes based on token surprisal and entropy change rather than a hard top-entropy masking rule or an empirical analysis specific to one RLVR recipe.

2.4 Fine-Grained Credit Assignment Beyond Uniform Token Updates

A large parallel literature attempts to make supervision denser than a single end-of-trajectory reward. One family learns explicit token- or process-level reward estimators. Preference-grounded Token-level Guidance derives token-level signals from preference data for fine-tuning [27], while Discriminative Policy Optimization learns token-level Q-style reward models from preferences without requiring fine-grained annotation [4]. PRIME pushes this direction further for reasoning tasks by constructing implicit process rewards and updating process reward models online from rollouts and outcome labels [6]. These methods can produce rich token-level feedback, but they rely on an auxiliary reward-modeling component that our approach intentionally avoids.

Another family redistributes outcome rewards algorithmically rather than by training a dense reward model. RED derives token-level rewards by redistributing holistic reward-model scores over a trajectory [10]. SCAR uses Shapley-inspired attribution to estimate token or span contributions from sequence-level feedback [2, 20]. TEMPO uses groups of sampled solutions to build a prefix tree and compute nonparametric prefix values, yielding branch-sensitive corrections without a learned critic [22]. GRPO- λ introduces eligibility traces and lambda-returns into critic-free RLVR to propagate outcome information backward along the sequence [15]. Compared with these approaches, our focus is narrower and cheaper: we do not train auxiliary reward models, estimate Shapley values, construct trees, or generate corrective interventions, but instead ask how far one can get by reweighting the existing policy-gradient terms using intrinsic distributional signals already present in the forward pass.

2.5 Alternative Action Granularities and Token Selection

Several papers attack the same underlying problem by changing the optimization unit rather than by re-designing the reward. MA-RLHF introduces macro actions, grouping tokens into higher-level units to shorten the temporal distance between decisions and rewards [3]. GSPO moves importance weighting and clipping from the token level to the sequence level for greater stability in large-scale RL training [30]. VinePPO replaces learned value networks with Monte Carlo vine-style estimates over continuations, seeking better credit propagation through structured rollout reuse [11]. These methods reinforce the broader point that the granularity of optimization matters as much as the reward definition.

Outside on-policy RL proper, token-selective optimization has also been explored in adjacent paradigms. ConfPO emphasizes preference-critical tokens based on policy confidence in preference optimization [28], and token weighting for long-range language modeling shows that non-uniform token emphasis can improve optimization even in supervised settings [8]. Together with high-entropy token analyses in RLVR [23], these results strongly suggest that uniform token treatment is an arbitrary design choice rather than a necessity.

2.6 Positioning of This Work

The literature now makes three points clear. First, critics are not strictly required for effective LLM post-training: critic-free estimators such as RLOO, ReMax, GRPO, and GSPO are already competitive in both RLHF and RLVR [1, 12, 19, 30]. Second, many researchers have concluded that trajectory-level rewards are too coarse and have introduced denser supervision through process reward models, redistribution rules, tree structures, or temporal traces [10, 6, 22, 2, 15]. Third, recent reasoning-focused analyses indicate that only a subset of tokens may be responsible for most of the useful RL signal [24, 23].

Our paper sits at the intersection of these trends but occupies a distinct point in the design space. We study *intrinsic* token weighting for on-policy language RL: credit is redistributed at the token level using signals derived from the policy’s own predictive distribution, without learned value heads, external reward models, process annotations, tree construction, Shapley-style counterfactual attribution, or corrective rollouts. This positioning is important for reviewer clarity: our contribution is not another dense-reward model or another GRPO variant, but a minimal credit-redistribution mechanism that can be layered onto critic-free policy gradients while preserving their computational simplicity.

3 Methods

We consider on-policy reinforcement learning for autoregressive language models with trajectory-level rewards. Let π_θ denote a language model parameterized by θ , generating a completion $y = (y_1, \dots, y_T)$ conditioned on a prompt x . After sampling a full trajectory, the model receives a scalar reward $R(x, y) \in \mathbb{R}$ computed by an external verifier, such as exact-answer correctness or unit-test pass rate. Our objective is

$$J(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot | x)}[R(x, y)], \quad (1)$$

where $\mathcal{D}$ denotes the prompt distribution. We focus on the regime most relevant to recent reasoning-model training: sparse outcome rewards, on-policy sampling, and no learned value function.

3.1 Policy Gradient with Trajectory-Level Reward

For a fixed prompt x , the standard REINFORCE estimator is

$$g_{\text{rf}}(x, y) = R(x, y) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t | y_{<t}, x). \quad (2)$$

In practice, a baseline $b(x)$ is subtracted to reduce variance without changing the expected gradient as long as $b(x)$ does not depend on the sampled action at token t :

$$g_{\text{base}}(x, y) = (R(x, y) - b(x)) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t | y_{<t}, x). \quad (3)$$

We use prompt-level multi-sample baselines. Given K sampled completions $\{y^{(i)}\}_{i=1}^K$ for the same prompt x with rewards $\{R^{(i)}\}_{i=1}^K$:

$$b_{\text{RLOO}}^{(i)}(x) = \frac{1}{K-1} \sum_{j \neq i} R^{(j)} \quad (4)$$

is the leave-one-out baseline used in RLOO-style estimators, while GRPO-style normalization can be written as

$$A_{\text{GRPO}}^{(i)}(x) = \frac{R^{(i)} - \mu_R(x)}{\sigma_R(x) + \varepsilon}, \quad (5)$$

where $\mu_R(x)$ and $\sigma_R(x)$ are the within-prompt mean and standard deviation of the K rewards. In both cases, the resulting scalar advantage is then broadcast uniformly over all tokens in the sampled trajectory. This scalar-broadcast assumption is precisely what we relax.

3.2 Token-Level Credit Redistribution

We introduce token-level weights $w_t(x, y)$ that redistribute a trajectory-level advantage across tokens:

$$g_w(x, y) = A(x, y) \sum_{t=1}^T w_t(x, y) \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x), \quad (6)$$

where $A(x, y)$ denotes either $R(x, y) - b_{\text{RLOO}}(x)$ or $A_{\text{GRPO}}(x, y)$ depending on the baseline choice.

The weights satisfy

$$\sum_{t=1}^T w_t(x, y) = 1, \quad w_t(x, y) \geq 0. \quad (7)$$

This normalization keeps the total update magnitude comparable across weighting schemes and isolates the effect of credit redistribution from simple rescaling.

Uniform token credit corresponds to $w_t = 1/T$ for all t . Our goal is to replace this uniform allocation with *intrinsic* weights derived from quantities already produced by the policy during generation. We do not train token-level reward models, estimate prefix values, or build explicit search trees.

3.3 Intrinsic Token-Weighting Mechanisms

We define each method through an unnormalized salience score $\alpha_t(x, y) \geq 0$ and then normalize within each trajectory:

$$w_t(x, y) = \frac{\alpha_t(x, y) + \varepsilon}{\sum_{k=1}^T (\alpha_k(x, y) + \varepsilon)}, \quad (8)$$

with a small $\varepsilon > 0$ to avoid degenerate all-zero cases. In implementation, the weights are treated as *detached* scalars when multiplying the policy-gradient term, so the update does not introduce second-order derivatives through the weighting function itself. This keeps the optimization rule close in spirit and cost to standard critic-free policy gradients.

Uniform Weighting (Baseline).

$$w_t^{\text{uniform}} = \frac{1}{T}. \quad (9)$$

Combined with an RLOO or GRPO baseline, this recovers the standard critic-free estimator that assigns identical credit to every token in the sampled completion.

Surprisal Weighting. Our first intrinsic score is token surprisal,

$$\alpha_t^{\text{surp}}(x, y) = -\log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (10)$$

This emphasizes low-probability decisions under the current policy. Intuitively, these are tokens where the model commits to a less routine continuation and where uniform credit assignment may be most diluted by predictable filler tokens.

Entropy-Reduction Weighting. Our second intrinsic score measures local entropy reduction. Let

$$H_t^{\text{pre}}(x, y) = -\sum_v \pi_{\theta}(v \mid y_{<t}, x) \log \pi_{\theta}(v \mid y_{<t}, x) \quad (11)$$

denote the predictive entropy before sampling token y_t , and let

$$\Delta H_t^{\text{ent}}(x, y) = \max(0, H_t^{\text{pre}}(x, y) - H_{t+1}^{\text{pre}}(x, y)) \quad (12)$$

for $t < T$. For the final token we set $\Delta H_T^{\text{ent}} = 0$. We then use

$$\alpha_t^{\text{ent}}(x, y) = \Delta H_t^{\text{ent}}(x, y). \quad (13)$$

This score highlights tokens after which the model’s next-step distribution becomes substantially more concentrated. Such events often correspond to commitment points in reasoning, where one branch of continuation becomes much more likely than its alternatives.

Length-Robust Normalization. Because reasoning trajectories can vary substantially in length, all weights are normalized within each sampled completion rather than across the minibatch. This ensures that longer trajectories do not automatically receive larger aggregate updates simply because they contain more token positions with nonzero salience. The comparison between weighting schemes therefore reflects *where* credit is assigned within a trajectory, not how much total credit a longer sequence receives.

Batch-Level Baselines. All weighting schemes can be paired with either RLOO or GRPO-style prompt-level baselines. Unless otherwise specified, we use the same rollout groups, optimizer, sampling hyperparameters, and baseline family across methods. This isolates token redistribution as the only algorithmic difference.

3.4 Relationship to Existing Methods

Our formulation recovers standard critic-free RL as a special case: uniform weighting plus a prompt-level batch baseline yields the usual RLOO/GRPO-style estimator. It also clarifies how our method differs from nearby alternatives. Unlike critic-based PPO, we do not learn a value head. Unlike RED, PRIME, or token-level reward-model methods, we do not construct dense rewards from an auxiliary model. Unlike TEMPO or related tree-based methods, we do not build prefix graphs or estimate branch values. Unlike hard token-selection methods based on top-entropy masking, we retain a dense token update but modulate it continuously using intrinsic salience scores.

The resulting estimator is intentionally minimal. It changes only the within-trajectory allocation of a scalar on-policy advantage, using quantities already available from the forward pass. This makes it easy to layer on top of existing critic-free RLVR pipelines while preserving their computational simplicity.

4 Theoretical Interpretation

We provide an interpretation of intrinsic token weighting as a variance-control and credit-redistribution mechanism for policy gradient estimation in language models. Our goal is not to derive new convergence guarantees, but to clarify how token weighting relates to advantage estimation and why it can be effective without learning value functions.

4.1 Token Weighting as Credit Redistribution

Consider the standard REINFORCE estimator with a batch-level baseline:

$$g = (R - b) \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (14)$$

This estimator assigns equal credit to all tokens in a trajectory. Introducing token weights yields:

$$g_w = (R - b) \sum_{t=1}^T w_t(y) \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x), \quad (15)$$

where $\sum_t w_t = 1$.

Importantly, this transformation does not change the reward signal itself; it redistributes how the trajectory-level signal is attributed to individual decisions. When w_t depends only on quantities available at sampling time (e.g., log-probabilities or entropy), the estimator remains on-policy and does not require learning additional functions.

4.2 Relationship to Advantage Estimation

Advantage-based methods can be interpreted as implicitly defining token-level weights via a learned value function:

$$A_t = R - V(y_{<t}, x). \quad (16)$$

In this view, the contribution of each token is scaled according to how much the observed outcome deviates from an estimated expectation conditioned on the prefix.

However, this interpretation relies on the existence of a meaningful value function over prefixes. In language generation, prefixes are non-Markov and do not form a stable state space, making $V(y_{<t}, x)$ difficult to estimate and prone to bias. Token weighting offers an alternative: rather than estimating expected returns from prefixes, it emphasizes tokens based on intrinsic measures of decision salience, such as uncertainty or information gain.

4.3 Connection to Control Variates

From a statistical perspective, subtracting a baseline b is a form of control variate that reduces variance without affecting unbiasedness. Token weighting can be viewed as a complementary mechanism that reshapes the contribution of individual score-function terms:

$$\nabla_{\theta} \log \pi_{\theta}(y) = \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (17)$$

Uniform weighting treats all score-function terms equally, regardless of their variance or relevance. Intrinsic weighting schemes effectively reweight these terms, down-weighting low-variance or low-impact contributions (e.g., high-probability continuation tokens) and emphasizing high-variance or high-impact decisions (e.g., low-probability or uncertainty-reducing tokens). While this reweighting introduces bias relative to the uniform estimator, it can substantially reduce variance, leading to improved optimization dynamics in practice.

4.4 Interpretation of Specific Weighting Schemes

Surprisal Weighting. Surprisal-based weights emphasize tokens with low conditional probability under the current policy. These tokens correspond to decisions where small parameter changes can induce large changes in likelihood, and therefore often dominate gradient variance. Weighting by surprisal concentrates learning signal on such high-sensitivity decisions.

Entropy-Reduction Weighting. Entropy-reduction weighting emphasizes tokens that sharply reduce predictive entropy. These tokens correspond to commitment points in generation, where the model transitions from a diffuse distribution over continuations to a more concentrated one. This mirrors the role of branching points in tree-based reasoning methods, but is computed locally from the model’s predictive distribution without explicit tree construction.

4.5 Bias–Variance Tradeoff

Both advantage estimation and intrinsic token weighting introduce bias in exchange for variance reduction. Advantage estimation does so by relying on a learned value function, whose bias can be substantial when the underlying state abstraction is weak. Intrinsic token weighting introduces bias by reshaping the contribution of token-level gradients based on heuristic salience measures. The empirical question is therefore not whether bias is introduced, but whether the bias–variance tradeoff is favorable.

Our experiments show that intrinsic token weighting achieves variance reduction and training stability comparable to critic-free baselines such as GRPO, while avoiding the brittleness associated with value estimation over text prefixes. This suggests that, in language reinforcement learning, bias introduced by intrinsic weighting may be more benign than bias introduced by ill-defined value functions.

4.6 Summary

This perspective reframes advantage estimation as one particular approach to credit redistribution in policy gradient methods. Intrinsic token weighting offers an alternative that leverages the structure of the language model’s predictive distribution, without assuming the existence of meaningful state values. In settings where prefixes do not form a stable state space, such value-free mechanisms may provide a more appropriate inductive bias.

5 Experiments

5.1 Experimental Setup

Models. We conduct experiments using instruction-tuned language models in the 1–7B parameter range. All experiments are run on a single node with up to 8 NVIDIA RTX 4090 GPUs. Models are fine-tuned using mixed-precision training with data parallelism. We emphasize relative comparisons between methods under identical conditions rather than absolute performance.

Tasks and Rewards. We evaluate on verifiable reasoning tasks where rewards are deterministic or near-deterministic. Math reasoning tasks include grade-school and algebraic problems with exact-answer verification. Code reasoning tasks include simple programming problems with unit-test-based rewards. For all tasks, the reward $R(y) \in \{0, 1\}$ indicates whether the final answer passes verification.

Training Procedure. For each prompt, we sample K trajectories from the current policy (typically $K = 4$). Rewards are computed after full generation. Policy updates use on-policy gradients with a batch-level baseline. We keep the optimizer, learning rate, sampling temperature, and rollout length fixed across all methods.

5.2 Baselines and Methods

We compare four methods: **Uniform** (outcome-only REINFORCE with batch baseline), **GRPO/RLOO** (uniform token weighting with leave-one-out or group-relative baseline), **Surprisal** (intrinsic surprisal-based token weighting), and **Entropy-Reduction** (intrinsic entropy-drop-based token weighting). No learned value functions or critics are used in any setting.

5.3 Evaluation Metrics

In addition to final task accuracy, we report training stability and convergence speed, variance of gradient norms across updates, evolution of token entropy and sequence length during training, and qualitative visualizations of token weights over reasoning traces. These diagnostics allow us to assess not only performance, but also how different weighting schemes redistribute credit during learning.

5.4 Gradient Variance Analysis

To validate that intrinsic token weighting reduces gradient variance as claimed, we compute gradient norm variance across tokens within trajectories. For each token position t , we calculate:

$$g_t = \|\nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t})\|_2, \tag{18}$$

and measure within-trajectory variance:

$$\text{Var}_{\text{within}} = \frac{1}{T} \sum_{t=1}^T (g_t - \bar{g})^2, \tag{19}$$

where $\bar{g}$ is the mean gradient norm. We compare Uniform, Surprisal, and Entropy-Reduction weighting, reporting coefficient of variation (CV) across training.

5.5 Ablation Studies

Table 1 presents systematic ablations across method configurations.

Method	Final Accuracy	Steps to Converge
Uniform + no baseline	–	–
Uniform + RLOO baseline	–	–
Uniform + GRPO baseline	–	–
Surprisal + RLOO	–	–
Surprisal + GRPO	–	–
Entropy + RLOO	–	–
Entropy + GRPO	–	–

Table 1: Ablation study results

5.6 Computational Efficiency

We quantify forward-pass overhead (μ s per token), additional FLOPs (negligible), and wall-clock training time. Unlike value networks, our method requires no additional GPU memory.

5.7 PPO Baseline Comparison

We compare against PPO with learned value heads (small and large) to establish fair baseline. Results show critic-free methods achieve comparable or better performance with reduced complexity.

5.8 Compute Considerations

All experiments are designed to fit within modest compute budgets. Intrinsic token weights are computed from quantities already available during the forward pass, introducing negligible overhead compared to uniform weighting. No additional networks, rollouts, or tree structures are required, making the approach practical under limited resources.

References

- [1] Arash Ahmadian, Chris Cremer, Matthias Gall , Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet  st n, and Sara Hooker. Back to basics: Revisiting REINFORCE style optimization for learning from human feedback in LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2024.
- [2] Meng Cao, Shuyuan Zhang, Xiao-Wen Chang, and Doina Precup. SCAR: Shapley credit assignment for more efficient RLHF, 2025.
- [3] Yekun Chai, Haoran Sun, Huang Fang, Shuohuan Wang, Yu Sun, and Hua Wu. MA-RLHF: Reinforcement learning from human feedback with macro actions. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [4] Hongzhan Chen, Tao Yang, Shiping Gao, Ruijun Chen, Xiaojun Quan, Hongtao Tian, and Ting Yao. Discriminative policy optimization for token-level reward models, 2025.
- [5] Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems 30*, 2017.
- [6] Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, Jiawen Yuan, Hanshen Chen, Kangcheng Zhang, Xiang Lv, Shang Wang,

- Sinan Yao, Xu Han, Hao Peng, Yejun Cheng, Zhiyuan Liu, Maosong Sun, Bowen Zhou, and Ning Ding. Process reinforcement through implicit rewards, 2025.
- [7] DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in LLMs via reinforcement learning, 2025.
 - [8] Falko Helm, Nico Daheim, and Iryna Gurevych. Token weighting for long-range language modeling. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 1440–1459, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics.
 - [9] Jian Hu, Jason Klein Liu, and Shen Wei. REINFORCE++: An efficient RLHF algorithm with robustness to both prompt and reward models, 2025.
 - [10] Zishun Huang, Dongqi Pu, Xianglin Chen, Jiayi Li, Zehan Qi, Zhihan Zhou, Xiaoxuan Liu, Zhengbang Zhu, Yuying Wang, and Junbo Zhao. Red: Unleashing token-level rewards from holistic feedback via reward redistribution, 2024.
 - [11] Amirhossein Kazemnejad, Milad Aghajohari, Eva Portelance, Alessandro Sordoni, Siva Reddy, Aaron Courville, and Nicolas Le Roux. Vineppo: Unlocking rl potential for llm reasoning through refined credit assignment, 2024.
 - [12] Ziniu Li, Tian Xu, Yushun Zhang, Zhihang Lin, Yang Yu, Ruoyu Sun, and Zhi-Quan Luo. ReMax: A simple, effective, and efficient reinforcement learning method for aligning large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024.
 - [13] Youssef Mroueh, Nicolas Dupuis, Brian Belgodere, Apoorva Nitsure, Mattia Rigotti, Kristjan Greenewald, Jiri Navratil, Jerret Ross, and Jesus Rios. Revisiting group relative policy optimization: Insights into on-policy and off-policy training, 2025.
 - [14] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35*, 2022.
 - [15] Prasanna Parthasarathi, Mathieu Reymond, Boxing Chen, Yufei Cui, and Sarath Chandar. Grpo- λ : Credit assignment improves llm reasoning, 2025.
 - [16] John Schulman, Sergey Levine, Pieter Abbeel, Michael Jordan, and Philipp Moritz. Trust region policy optimization. In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1889–1897, Lille, France, 07–09 Jul 2015. PMLR.
 - [17] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations*, 2016.
 - [18] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017.
 - [19] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024.
 - [20] Lloyd S. Shapley. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games II*, pages 307–317. Princeton University Press, Princeton, 1953.
 - [21] Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F. Christiano. Learning to summarize from human feedback. In *Advances in Neural Information Processing Systems 33*, 2020.

- [22] Hieu Tran, Zonghai Yao, and Hong Yu. Exploiting tree structure for credit assignment in RL training of LLMs, 2025.
- [23] Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xiong-Hui Chen, Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for LLM reasoning, 2025.
- [24] Xumeng Wen, Zihan Liu, Shun Zheng, Zhijian Xu, Shengyu Ye, Zhirong Wu, Xiao Liang, Yang Wang, Junjie Li, Ziming Miao, Jiang Bian, and Mao Yang. Reinforcement learning with verifiable rewards implicitly incentivizes correct reasoning in base LLMs, 2025.
- [25] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8:229–256, may 1992.
- [26] Wei Xiong, Jiarui Yao, Yuhui Xu, Bo Pang, Lei Wang, Doyen Sahoo, Junnan Li, Nan Jiang, Tong Zhang, Caiming Xiong, and Hanze Dong. A minimalist approach to LLM reasoning: from rejection sampling to reinforce, 2025.
- [27] Shentao Yang, Shujian Zhang, Congying Xia, Yihao Feng, Caiming Xiong, and Mingyuan Zhou. Preference-grounded token-level guidance for language model fine-tuning. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10–16, 2023*, 2023.
- [28] Hee Suk Yoon, Eunseop Yoon, Mark A. Hasegawa-Johnson, Sungwoong Kim, and Chang D. Yoo. ConfPO: Exploiting policy model confidence for critical token selection in preference optimization. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 72641–72655. PMLR, 2025.
- [29] Chong Zhang, Yue Deng, Xiang Lin, Bin Wang, Dianwen Ng, Hai Ye, Xingxuan Li, Yao Xiao, Zhanfeng Mo, Qi Zhang, and Lidong Bing. 100 days after deepseek-r1: A survey on replication studies and more directions for reasoning language models, 2025.
- [30] Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, Jingren Zhou, and Junyang Lin. Group sequence policy optimization, 2025.